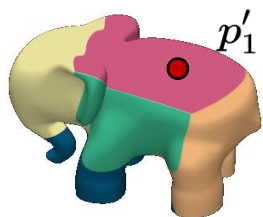
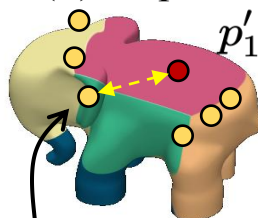


(a) Farthest-Point Sampling (Offline):

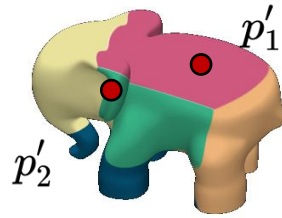
(i) Step 1: Initialization



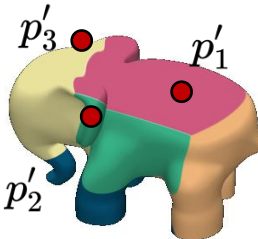
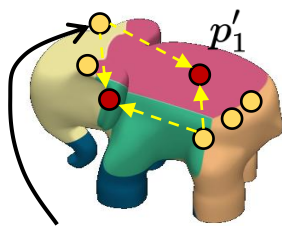
(ii) Step 2: Next Sample Generation



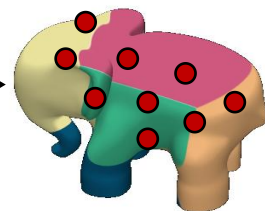
$$p'_2 = \arg \max_{p_i \in \mathcal{F}} \min_{p'_j \in \mathcal{I}_k} \|\mathbf{p}_i - \mathbf{p}'_j\|^2$$



$$\mathcal{I}_k = \{p'_1, p'_2\}$$



Repeated
Generation

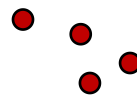
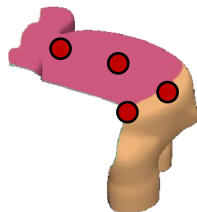
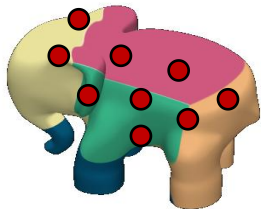


$$p'_3 = \arg \max_{p_i \in \mathcal{F}} \min_{p'_j \in \mathcal{I}_k} \|\mathbf{p}_i - \mathbf{p}'_j\|^2$$

$$\mathcal{I}_k = \{p'_1, p'_2, p'_3\}$$

$$\mathcal{I}_{\text{obj}} = \{p_i\}_{i=1}^{n_s}$$

(b) Valid Region Selection (Online):



$\mathcal{I}_{\text{obj}}$: Sampled Candidate Set

$\mathcal{F}_v$: Valid Surface Region

$\mathcal{I}_{\text{avail}}$: Valid Candidate Set